

Question 1 (14 marks)**Marks**

- (a) Evaluate $\frac{\sqrt{125 - 7.1^2}}{2009}$. Write your answer in scientific notation correct to two significant figures. **2**
- (b) Write the recurring decimal $0.0\dot{8}\dot{7}$ in the form $\frac{a}{b}$ where a and b are integers. **2**
- (c) Factorise $27 - n^3$ **1**
- (d) Express $\frac{3 + \sqrt{2}}{3 - \sqrt{2}}$ with a rational denominator in simplest form. **3**
- (e) Solve $|2x + 3| > 21$ **2**
- (f) Solve $2^{3x} = 32$ **2**
- (g) Find the exact solution(s) to $2x^2 = -7x - 3$ in simplest form. **2**

Question 2 (14 Marks) *Start this question on a new page.***Marks**

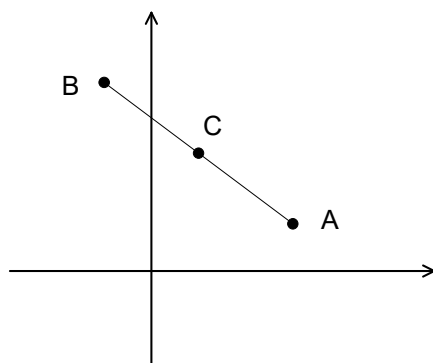
- (a) Consider the function $f(x) = x^2 + 6x + 8$.
- (i) Find the vertex. **2**
- (ii) Find the y-intercept. **1**
- (iii) Find the x-intercepts. **2**
- (iv) Draw a neat sketch showing the vertex and any intercepts. **1**
- (v) State the domain and range of the function. **2**
- (b) Show that $f(x) = x^5 + 3x^3$ is an odd function. **2**
- (c) Draw a neat sketch of $y = \frac{1}{x+1}$ **2**
- (d) Draw the region defined by $x^2 + y^2 < 4$ **2**

Question 3 (14 Marks) *Start this question on a new page.* **Marks**

- (a) Find the exact value of $\cos 150^\circ$ **1**
- (b) Solve $2 \sin \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$ **2**
- (c) Prove that $\frac{1 - \sin^2 \theta}{1 - \cos^2 \theta} = \cot^2 \theta$ **2**
- (d) If $\cos \theta = \frac{5}{6}$ and $\sin \theta < 0$ then find the exact value of $\tan \theta$. **3**
- (e) A hiker leaves base camp on a bearing of 48° and travels for 2.4 km until he hits the highway. The highway runs directly North-South, and he follows it South for 2.2 km.
- (i) Draw a neat diagram clearly indicating all relevant information. **1**
- (ii) Show that the distance (as the crow flies) that the hiker is from the base camp is 1.88 km (to 2 decimal places). **2**
- (iii) What bearing is the base camp from his current location? **3**
Give your answer to the nearest degree.

Question 4 (14 Marks) *Start this question on a new page.* **Marks**

- (a) In the diagram below A is the point (3,1) and B is the point (-1,7).



NOT TO SCALE

- (i) Find the exact length of the interval AB. **2**
- (ii) Find the gradient of the interval AB. **1**
- (iii) Find the coordinates of C, the midpoint of the interval AB. **1**
- (iv) Find the equation of the line that passes through C and is perpendicular to AB. Give your answer in general form. **3**

Question 4 continued.

- (b) Find the equation of the line that makes an angle of 60° with the positive x -axis and has y -intercept of -3 . Write your equation in the form $y = mx + b$ where m and b are exact values. **2**

- (c) The two lines below intersect at the point P.

$$2x - y + 1 = 0$$

$$3x + y - 11 = 0$$

- (i) Show that the coordinates of P is $(2, 5)$ **2**
- (ii) Find the equation of the line that passes through P and the point $(3, 1)$. Give your answer in general form. **3**

Question 5 (14 Marks) *Start this question on a new page.***Marks**

- (a) Differentiate:

(i) $y = 9x^5$ **1**

(ii) $f(x) = (2x + 1)(x - 2)^5$ **2**

(iii) $y = \frac{x^2}{x^2 + 1}$ **3**

- (b) The derivative of a function is defined as $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ **3**

Use this equation to differentiate $f(x) = x^2 - 1$

- (c) For the curve $y = x^3 - 3x^2 + 3x - 1$

- (i) Find the gradient of the tangent to the curve at $x = 2$. **2**

- (ii) Hence find the equation of the tangent to the curve at $x = 2$. **3**
Give your answer in general form.

Question 6 (14 Marks) *Start this question on a new page.***Marks**

- (a) If the two roots of the equation $y = 2x^2 - 6x + 5$ are α and β then find the value of:
- | | |
|----------------------------|----------|
| (i) $\alpha + \beta$ | 1 |
| (ii) $\alpha\beta$ | 1 |
| (iii) $\alpha^2 + \beta^2$ | 2 |
- (b) Find the values of k for which $x^2 + 4kx + 16 = 0$ has equal roots. **3**
- (c) Solve the inequation $q^2 \leq 3q + 40$. **3**
- (d) Find the equation of the parabola that passes through the points $(0,5)$, $(2,3)$ and $(4,9)$. **4**

END OF PAPER

Question 1

(a) 0.004298929
 $= 4.3 \times 10^{-3}$

(b) let $x = 0.0878787\ldots$
 $100x = 8.7878787\ldots$

$\therefore 99x = 8.7$

$\therefore x = 8.7/99$
 $= 87/990$
 $= 29/330$

(c) $27 - n^3 = (3 - n)(9 + 3n + n^2)$

(d) $\frac{3+\sqrt{2}}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}} = \frac{9+6\sqrt{2}+2}{9-2}$
 $= \frac{11+6\sqrt{2}}{7}$

(e) if $2x+3 > 0$ then:

$2x+3 > 21$

$2x > 18$

$x > 9$

if $2x+3 < 0$ then:

$-2x-3 > 21$

$\therefore -2x > 24$

$x < -12$

Hence $x < -12$ or $x > 9$

(f) $2^{3x} = 2^5$

$3x = 5$

$x = 5/3$

(g) $2x^2 + 7x + 3 = 0$

$(2x+1)(x+3) = 0$

$x = -1/2$ or -3

QUESTION 2

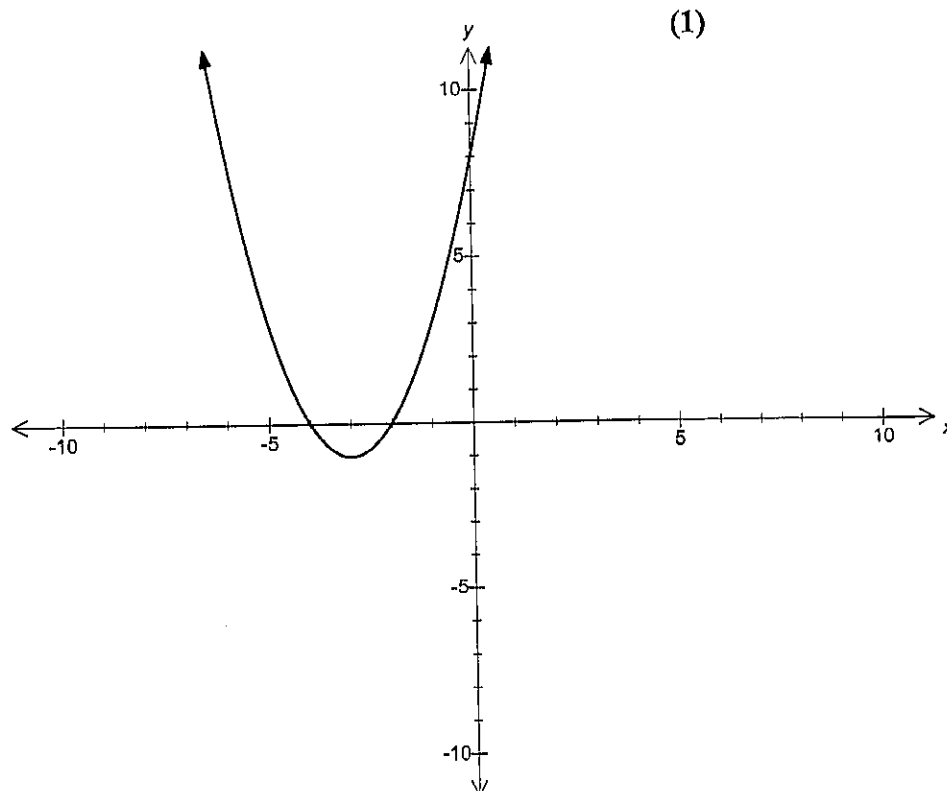
a) i) $x = \frac{-b}{2a}$ $\therefore y = (-3)^2 + 6(-3) + 8$ $\therefore$ Vertex $(-3, -1)$

$x = \frac{-6}{2}$ $y = 9 - 18 + 8$
 $x = -3$ (1) $y = -1$ (1)

ii) y-int when $x = 0$
 $\therefore y = (0)^2 + 6(0) + 8$
 $y = 8$
 $\therefore (0, 8)$ (1)

iii) x-int when $y = 0$
 $x^2 + 6x + 8 = 0$
 $(x + 4)(x + 2) = 0$
 $x = -4$ or $x = -2$
 $\therefore (-4, 0)$ $(-2, 0)$ (2)

iv)



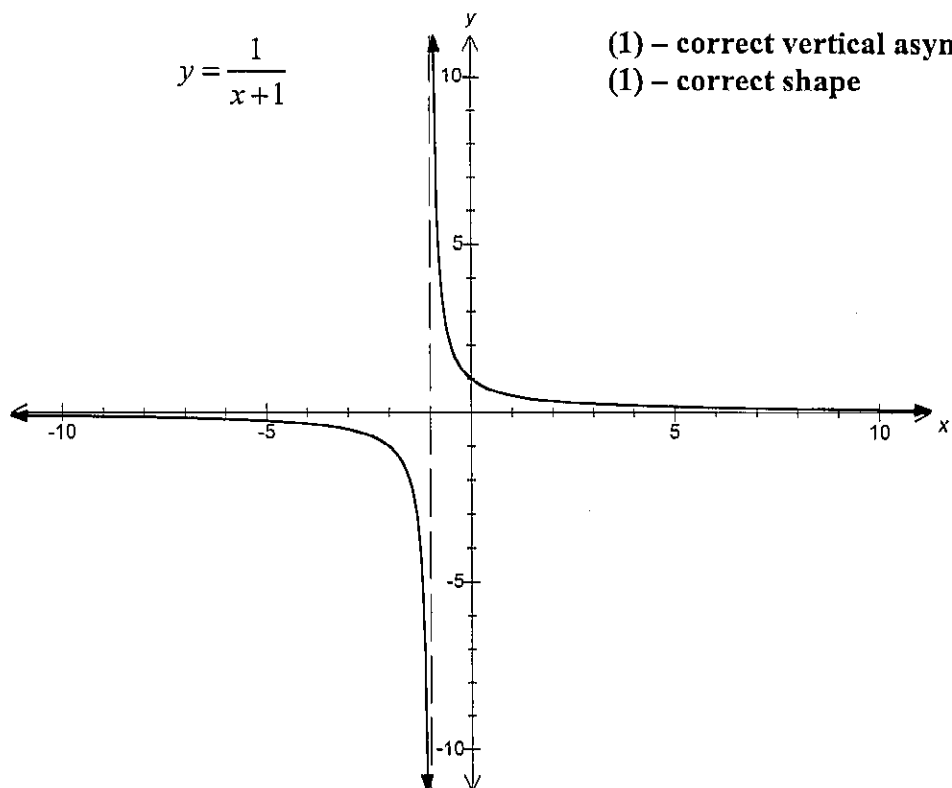
v) Domain: all real x-values (1)
 Range: $y \geq -1$ (1)

b) For an odd function $f(-x) = -f(x)$ given $f(x) = x^5 + 3x^3$ $\text{or } f(x) = -f(-x)$
 $f(-x) = (-x)^5 + (-x)^3$ and $-f(x) = -(x^5 + 3x^3)$
 $= -x^5 - x^3$ (1) $= -x^5 - x^3$ (1)
 $\therefore$ Function is an odd function.

If show $f(x) \neq f(-x)$ and (1)

c)

$$y = \frac{1}{x+1}$$

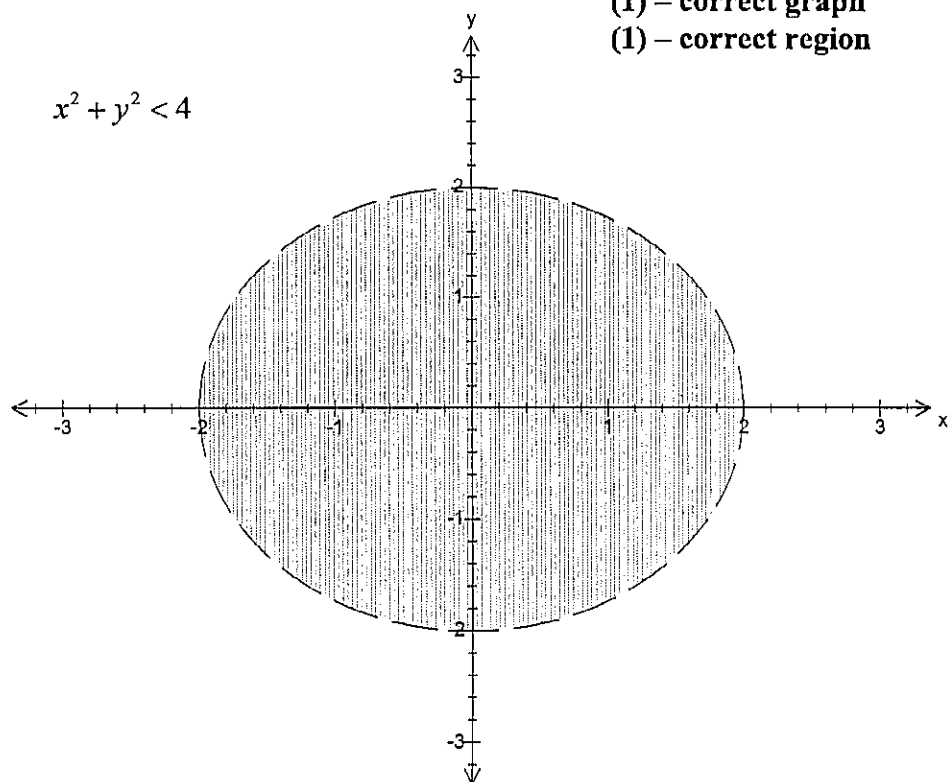


(1) – correct vertical asymptote

(1) – correct shape

d)

$$x^2 + y^2 < 4$$



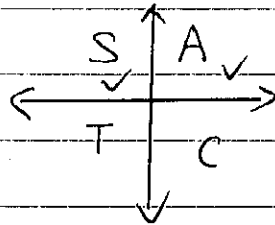
(1) – correct graph

(1) – correct region

Year 11 Math Preliminary 2009 Sample Answers Question 3

(a) $\cos(150^\circ) = \cos(180^\circ - 30^\circ) = -\cos 30^\circ = -\frac{\sqrt{3}}{2}$ ①

(b) $2\sin\theta = 1$
 $\sin\theta = \frac{1}{2}$
 $\theta = 30^\circ$



$\sin 30^\circ = \sin(180^\circ - 150^\circ)$
 $= \sin 150^\circ$
 $= \frac{1}{2}$

$\therefore \theta = 30^\circ, 150^\circ$
 ① ①

①

(c) $\cos^2\theta + \sin^2\theta = 1 \Rightarrow \cos^2 = 1 - \sin^2\theta$ and $\sin^2\theta = 1 - \cos^2\theta$

LHS = $\frac{1 - \sin^2\theta}{1 - \cos^2\theta}$

= $\frac{\cos^2\theta}{\sin^2\theta}$

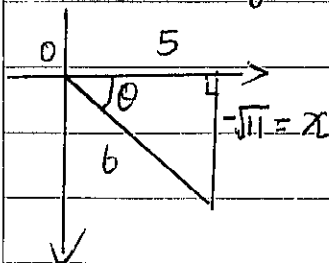
= $\left(\frac{\cos\theta}{\sin\theta}\right)^2$

= $(\cot\theta)^2$

= $\cot^2\theta$ ①

= RHS

(D) $\cos\theta = \frac{5}{6}$ $x^2 + 5^2 = 6^2$



$x^2 = 36 - 25$

$x^2 = 11$

$x = \pm\sqrt{11}$ ①

$\tan\theta = \frac{\sin\theta}{\cos\theta}$

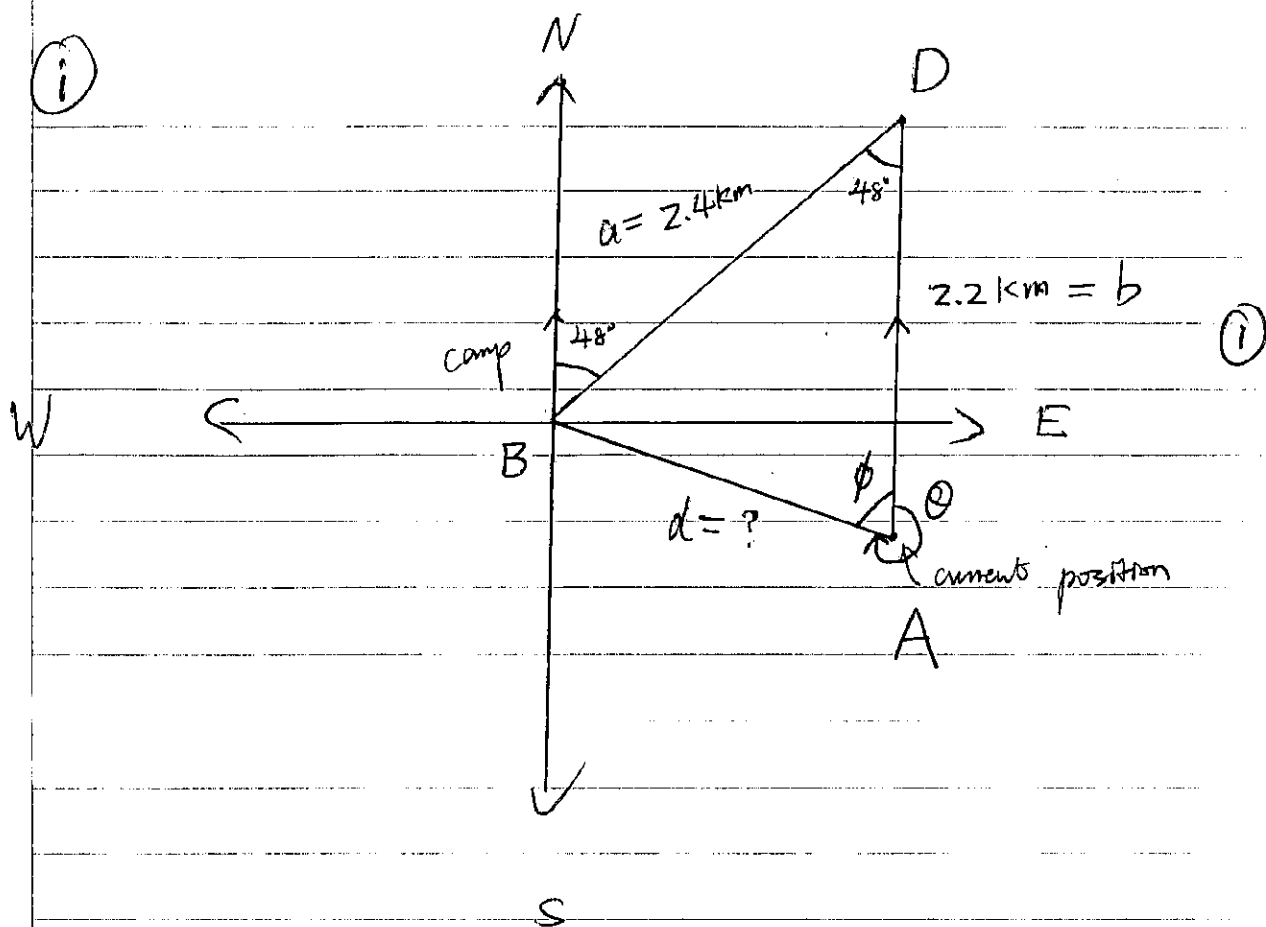
given $\cos\theta > 0$ and $\sin\theta < 0$

$\therefore \tan\theta < 0 \Rightarrow x = -\sqrt{11}$

$\therefore \tan\theta = \frac{-\sqrt{11}}{5} = -\frac{\sqrt{11}}{5}$

①

(E) (i)



(ii) $\angle NBD = \angle BDA$ (Alternate angles, $BN \parallel AD$)

$$d^2 = a^2 + b^2 - 2ab \cos(\angle BDA) \quad \text{①} \quad \text{Cosine Rule}$$

$$= (2.4)^2 + (2.2)^2 - (2)(2.4)(2.2)(\cos 48^\circ)$$

$$= 3.53398$$

$$\sqrt{d^2} = d = 1.88 \text{ km} \quad \text{①} \quad \text{to 2 decimal places}$$

(iii) $\frac{\sin \phi}{a} = \frac{\sin \angle BDA}{d} \quad \text{①} \quad \text{Sine Rule}$

$$\frac{\sin \phi}{2.4} = \frac{\sin 48^\circ}{1.88}$$

$$\sin \phi = \frac{\sin 48^\circ \times 2.4}{1.88}$$

$$\sin^{-1}\left(\frac{\sin 48^\circ \times 2.4}{1.88}\right) = \phi$$

$$\phi = 71^\circ 34' \quad \text{①}$$

$$\theta = 360^\circ - 71^\circ 34'$$

$$= 288^\circ 26'$$

$$= 288^\circ \text{ (To nearest degree.)}$$

$\therefore$ Bearing of camp
from current position is
① 288° T

Question 4 Year 11 Mathematics Preliminary 2009.

$$\begin{aligned} (a)(i) AB &= \sqrt{(3-1)^2 + (1-7)^2} \\ &= \sqrt{16 + 36} \quad \checkmark \\ &= \sqrt{52} \quad \checkmark \\ &= 2\sqrt{13} \end{aligned}$$

(2) [If they used the calculator a mark was deducted]

(ii) Equation of line through (2, 5) and (3, 1)

$$m = \frac{1-5}{3-2}$$

$$m = -4 \quad \checkmark$$

$$y - 1 = -4(x - 3) \quad \checkmark$$

$$y - 1 = -4x + 12$$

$$\therefore 4x + y - 13 = 0 \quad \checkmark \quad (3)$$

$$\begin{aligned} (iii) M_{AB} &= \frac{7-1}{-1-3} \\ &= \frac{6}{-4} \\ &= -\frac{3}{2} \quad \checkmark \quad (1) \end{aligned}$$

$$(iii) \text{Midpoint } AB = \left(\frac{3+1}{2}, \frac{1+7}{2} \right)$$

$$\therefore C = (1, 4) \quad \checkmark \quad (1)$$

[They need brackets for the point but a mark was not deducted]

(iv) Equation of line

$$m = \frac{2}{3} \quad \checkmark \quad C = (1, 4)$$

$$y - 4 = \frac{2}{3}(x - 1) \quad \checkmark$$

$$3y - 12 = 2x - 2$$

$$0 = 2x - 3y + 10 \quad \checkmark \quad (3)$$

$$(b) \tan 60^\circ = m$$

$$\therefore m = \sqrt{3} \quad \checkmark$$

$$\therefore y = \sqrt{3}x - 3 \quad \checkmark \quad (2)$$

[1 mark for $\sqrt{3}$ or -3 correctly used.
1 mark for $y = mx + b$ in this order only]

$$(c) 2x - y + 1 = 0 \quad \text{--- (1)}$$

$$3x + y - 11 = 0 \quad \text{--- (2)}$$

$$\textcircled{1} + \textcircled{2} \quad 5x - 10 = 0$$

$$5x = 10$$

$$\therefore x = 2 \quad \checkmark$$

$$\text{Sub in } \textcircled{1} \quad 4 - y + 1 = 0$$

$$5 - y = 0$$

$$y = 5 \quad \checkmark$$

$$\therefore P \text{ is } (2, 5)$$

[Both parts $x=2, y=5$ must be shown.
1 mark each part.]

Question 5

a) 1. $y = 9x^5$

$$\frac{dy}{dx} = 45x^4$$

①

(ii) $f(x) = (2x+1)(x-2)^5$
 $= uv$

$$u = 2x+1$$

$$\frac{du}{dx} = 2$$

$$v = (x-2)^5$$

$$\frac{dv}{dx} = 5(x-2)^4$$

$$f'(x) = v \frac{du}{dx} + u \frac{dv}{dx}$$

$$= 2(x-2)^5 + (2x+1)5(x-2)^4$$

$$= (x-2)^4 [2(x-2) + 5(2x+1)]$$

$$= (x-2)^4 (12x+1)$$

① mark
 ignore subsequent
 errors

(iii) $y = \frac{x^2}{x^2+1}$
 $= \frac{u}{v}$

$$u = x^2$$

$$\frac{du}{dx} = 2x$$

$$v = x^2+1$$

$$\frac{dv}{dx} = 2x$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$= \frac{2x(x^2+1) - 2x \times x^2}{(x^2+1)^2}$$

$$= \frac{2x^3 + 2x - 2x^3}{(x^2+1)^2}$$

$$= \frac{2x}{(x^2+1)^2}$$

①

* many students
 have not learnt
 the product
 or quotient rules

$$(b) f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 - 1 - (x^2 - 1)}{h} \quad (1)$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2hx + h^2 - 1 - x^2 + 1}{h} \quad (1)$$

$$= \lim_{h \rightarrow 0} \frac{h(2x+h)}{h}$$

$$= 2x \quad (1)$$

no marks if they just wrote $f'(x) = 2x$

$$c) (i) y = x^3 - 3x^2 + 3x - 1$$

$$\frac{dy}{dx} = 3x^2 - 6x + 3 \quad \text{at } x=2 \quad (1)$$

$$= 3(2)^2 - 6(2) + 3$$

$$= 12 - 12 + 3$$

$$= 3 \quad (1)$$

$\therefore$ gradient of the tangent at $x=2$ is 3

$$(ii) \text{ when } x=2, y = x^3 - 3x^2 + 3x - 1$$

$$= 2^3 - 3(2)^2 + 3(2) - 1$$

$$= 1 \quad (1)$$

some students have done this in part (i).

$$y - y_1 = m(x - x_1) \quad m=3, (2,1)$$

$$y - 1 = 3(x - 2) \quad (1)$$

$$y - 1 = 3x - 6$$

$$3x - y - 5 = 0 \quad (1)$$

Marks Solutions

Q6 (a) $y = 2x^2 - 6x + 5$

(i) $\alpha + \beta = -\frac{b}{a} = -\frac{(-6)}{2} = 3$. (1)

(ii) $\alpha\beta = \frac{c}{a} = \frac{5}{2}$. (1)

(iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$. 1 mark identity (2)
 $= 3^2 - 2(\frac{5}{2})$. (3)
 $= 4$. 1 mark (1)

(b) $x^2 + 4kx + 16 = 0$ has equal roots if $\Delta = 0$ (2)

$\therefore (4k)^2 - 4(1)(16) = 0$ 1 mark equating to zero.

$\therefore 16k^2 - 64 = 0$

$\therefore 4k^2 - 4 = 0$

$\therefore (k-2)(k+2) = 0$ 1 mark factorisation (3)

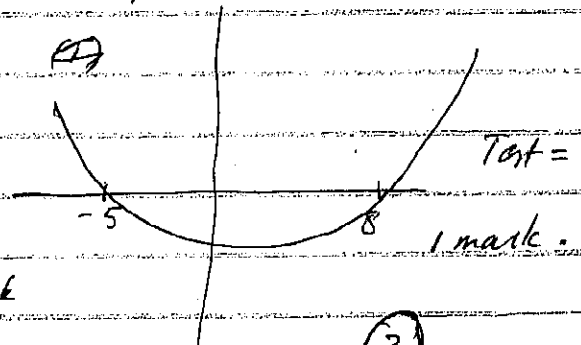
$\therefore k = 2$ or -2 . (2)
 1 mark solution

(c) $q^2 \leq 3q + 40$

$q^2 - 3q - 40 \leq 0$ 1 mark

$\therefore (q-8)(q+5) \leq 0$ 1 mark

$-5 \leq q \leq 8$. 1 mark. (3)



(d) $y = ax^2 + bx + c$ if $x = 0$ $y = 5$ 1 mark.
 $\therefore c = 5$

$\therefore y = ax^2 + bx + 5$ if $x = 2$ $y = 3$.

$3 = 4a + 2b + 5$

$\therefore 4a + 2b = -2$

$\therefore 2a + b = -1$ (1) 1 mark

if $x = 4$ $y = 9$

$\therefore 9 = 16a + 4b + 5$

$\therefore 16a + 4b = 4$

$\therefore 4a + b = 1$ (2) 1 mark

$$\therefore 2a = 2$$

$$\therefore a = 1.$$

$$\therefore b = -3$$

$$\therefore y = x^2 - 3x + 5. \quad 1 \text{ mark.}$$